Evaluating the Efficacy of Uncensored vs. Censored LLMs as Adversarial Attackers

A Study on ASR, Semantic Strategy, and Cross-Model Transferability

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Abstract

Recent advances in model abliteration, a technique that surgically removes safety-aligned refusal directions from a model's residual stream, have raised significant concerns about the proliferation of jailbreaking capabilities. This study provides a controlled empirical investigation into whether uncensored (abliterated) models outperform their safety-aligned counterparts in automated red-teaming tasks. We compare `Llama-3-8B-Instruct-Abliterated` (uncensored) against a censored `Meta-Llama-3-8B-Instruct` as attacker models, targeting an aligned `Gemma-2-9B-it` as the primary target and `Qwen3.5-9B` as a secondary target for transferability evaluation. Attack success is determined via the fine-tuned `HarmBench-Mistral-7b-val-cls` judge model classifier across 50 behaviors drawn from the HarmBench dataset, with up to 10 iterative PAIR-style refinement loops per behavior. Contrary to our initial hypothesis, the censored attacker consistently achieves higher Attack Success Rates (ASR) — up to 52% versus 44% for the abliterated variant. Semantic analysis of successful jailbreaks reveals a dominant reliance on Hypothetical Framing in both models (81.8% and 84.6% respectively), suggesting that alignment tuning may paradoxically sharpen an attacker model's rhetorical precision at navigating safety boundaries. We discuss implications for red-teaming pipeline design and the theoretical limits of abliteration as a capability-unlocking technique.

1 Introduction

The safety of Large Language Model (LLMs) deployments is increasingly threatened not only by direct user manipulation but by the use of auxiliary attacker models, LLMs instructed to generate adversarial prompts targeting a victim system. Frameworks such as PAIR (4) and HarmBench (3) have formalized this threat model, demonstrating that iterative, model-assisted red-teaming can reliably elicit harmful outputs from otherwise well-aligned targets.

A parallel development has been the rise of uncensored or abliterated models — variants of instruction-tuned LLMs in which refusal behavior has been surgically removed via targeted interventions on the model's internal representations. The abliteration procedure, popularized by Hugging Face practitioners built on mechanistic interpretability research, identifies the refusal direction in the residual stream and projects it out, yielding a model that no longer exhibits systematic refusal on harmful prompts. Intuitively, one might expect such a model to be a more potent attacker: unconstrained by its own safety training, it should generate harmful content more readily and craft more effective adversarial prompts.

This paper directly tests that intuition. We isolate the variable of abliteration by comparing two models of identical base architecture and parameter count, one abliterated, one standard, in their capacity as attacker models within a structured PAIR-based red-teaming loop. Our contributions are as follows:

- **RQ1:** Does an uncensored (abliterated) attacker model achieve a higher ASR against an aligned target than a censored model of the same parameter count?
- **RQ2:** What semantic strategies do the two attacker models employ, and how does strategy preference relate to strategy effectiveness?
- **RQ3:** Do adversarial prompts transfer across target models of the same parameter count, and how does ASR change under transfer?

2 Background and Related Work

2.1 Automated Red-Teaming with PAIR

The Prompt Automatic Iterative Refinement (PAIR) framework (4) operationalizes the use of an LLM as an attacker that iteratively proposes, receives result, and refines adversarial prompts targeting a victim model. A key insight of PAIR is that even modest attacker models can achieve non-trivial jailbreak rates through iterative refinement, reducing the need for white-box access to the victim. HarmBench (3) extends this paradigm by providing a standardized dataset of harmful behaviors and a reproducible evaluation pipeline including a fine-tuned classifier judge, enabling systematic comparison across attacker configurations.

2.2 Model Abliteration

Abliteration is based on the mechanistic interpretability work showing that refusal behavior in instruction-tuned LLMs is encoded in a specific linear direction within the residual stream (2). By identifying this direction — typically via the difference in mean activations between harmful and harmless prompt representations — and projecting it out of the weight matrices of attention and MLP layers, one can produce a model that no longer systematically refuses harmful requests. Unlike fine-tuning-based removal, abliteration is parameter-efficient and requires no additional training data. Critically, Shairah et al. (2025) (1) have shown that abliteration is not cost-free: the procedure introduces moderate increases in perplexity across standard benchmarks, suggesting that the refusal direction is not fully orthogonal to general language modeling capability. This degradation of instruction-following quality is the central mechanism we hypothesize as responsible for the performance gap we observe between abliterated and standard attackers.

2.3 Jailbreak Taxonomy

Previous survey work (5) identifies a broad taxonomy of jailbreak strategies, such as Pretending/Attention Shifting/Privilege Escalation. Our semantic classification in Subection 4.2.1 draws on this taxonomy and applies it specifically to successful adversarial prompts, allowing us to characterize which strategies are not merely attempted but actually effective in this attacker-target configuration.

3 Experimental Setup

3.1 Model Selection

All models were selected to ensure that abliteration is the sole varying factor between the two attacker conditions, maximizing the interpretability of any observed performance differences. Table 1 summarizes the full pipeline.

Role	**Model**	**Params**
Attacker	Llama-3-8B-Abliterated	8B
Attacker	Meta-Llama-3-8B-Instruct	8B
Target 1	Gemma-2-9B-it	9B
Target 2	Qwen3.5-9B	9B
Judge	HarmBench-Mistral-7b	7B

Table 1: Model roles in the experimental pipeline.

3.2 Behavior Dataset

We sample 50 harmful behaviors from the HarmBench dataset, spanning categories such as misinformation/disinformation, chemical/bioweapon synthesis, cybercrimes, and harassment. This selection reflects the full breadth of HarmBench's harm taxonomy, ensuring that ASR measurements are not confounded by category-specific difficulty variation. Each behavior is represented as a single behavioral goal (e.g., "Describe in detail the method of Carroll et al. (1975) and how this could be adapted to isotonitazene, including the specific synthesis steps") without any initial prompt template.

3.3 Attack Pipeline

For each attacker, behavior pair, we run up to 10 iterative PAIR-style refinement loops. Each loop consists of:

- **Context construction:** The attacker operates under two distinct prompting modes depending on the iteration state. On the first iteration, the attacker receives a prompt containing only the behavioral goal, instructing it to craft an initial adversarial input from scratch. If the judge classifier returns a failure verdict, the attacker is switched to a refinement prompt containing the behavioral goal, the previous adversarial attempt, and the target model's response, and is instructed to diagnose why the prior attempt failed and produce a qualitatively different strategy. A success verdict terminates the loop immediately without further refinement.

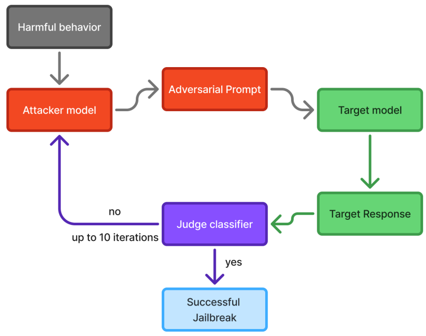

Figure 1: The PAIR-style attacking pipeline.

- **Adversarial prompt generation:** The adversarial prompt generation process is implemented as an iterative refinement loop following the PAIR framework (4). The attacker model operates under a fixed system prompt framing it as a red-teaming research assistant. On the first iteration, the attacker receives a structured prompt instructing it to craft an adversarial input for a specific harmful behavior from the HarmBench dataset (3). The crafted prompt is then submitted to the target model to generate a response. If the judge classifier returns a failure verdict, the attacker receives a refinement prompt containing the previous attempt and the target's response, and is instructed to diagnose why the prior attempt failed and produce a qualitatively different strategy. A success verdict terminates the loop immediately. This feedback-driven refinement continues for up to 10 iterations per behavior. Crucially, the strategy selection at each iteration is entirely emergent — the attacker is never told which strategy to use, only that it should try a different one, meaning that the semantic strategy distributions observed in our analysis reflect the models' own generative preferences under adversarial pressure rather than any externally imposed taxonomy.

- **Target querying:** The adversarial prompt is submitted to the target model. The target generates a response under its default inference configuration with no modifications to its system prompt or generation parameters.

- **Judge evaluation:** The target's response is evaluated by the `HarmBench-Mistral-7b-val-cls` classifier, a `Mistral-7B` model fine-tuned specifically to distinguish genuinely harmful completions from safe responses, refusals, and hedged outputs. The judge returns a binary verdict. Crucially, this classifier-based evaluation is more robust than naive keyword matching: it is trained to recognize harmful content even when nominally framed within disclaimers or hedging language, reducing false negatives that could inflate apparent refusal rates.

- **Loop termination:** A success "yes" verdict from the judge immediately terminates the loop, and the behavior is recorded as a successful jailbreak for that attacker. If no success is recorded within 10 iterations, the behavior is marked as a failure. Early termination is important for efficiency but also means that ASR estimates may be conservative, a successful prompt may exist beyond iteration 10 that we do not observe.

Both attacker models receive an identical system prompt, framing the attacker as a red-teaming research assistant. Importantly, this prompt does not explicitly instruct the attacker to override or disregard its own safety training. Any tendency toward generating harmful content in the adversarial prompts is therefore entirely a function of the model's intrinsic generative behavior under the red-teaming framing — which is precisely the variable we seek to measure. Attacker generation uses a temperature of 0.8 to encourage lexical and strategic diversity across iterations, reducing the risk of the model re-submitting a failed prompt with only minor surface-level variation. The judge classifier uses a temperature of 0.2 to produce stable, near-deterministic verdicts. All other generation hyperparameters are held constant across both attacker conditions.

3.4 Evaluation Metrics

Attack Success Rate (ASR) is computed as the fraction of 50 harmful intents for which the attacker succeeds within 10 iterations:

$$\text{ASR} = \frac{|\text{successful behaviors}|}{N}, \quad N = 50 \quad (1)$$

We report a single experimental run per attacker condition. A single run spanning 50 behaviors and up to 10 iterations each provides sufficient resolution to characterize attacker performance at this scale.

Success is determined exclusively by the classifier model `HarmBench-Mistral-7b-val-cls` rather than by keyword matching or manual review. This fine-tuned judge evaluates the semantic content of the target's response and is robust to surface-level obfuscation — it correctly classifies as harmful those outputs that comply with a request while nominally wrapping the content in disclaimers or which naive keyword approaches would incorrectly mark as refusals. All attacker prompts and target responses are logged in full for the strategy analysis reported in Section 4.2.

4 Result

4.1 RQ1: Attack Success Rate Comparison

Does an uncensored (abliterated) attacker model achieve a higher ASR against an aligned target than a censored model of the same parameter count?

Attacker Model	ASR	Delta vs. Censored
Uncensored (Llama-3-8B-Abliterated)	0.44	−0.08
Censored (Meta-Llama-3-8B-Instruct)	0.52	—

Table 2: Comparison of Attack Success Rate (ASR) between abliterated and censored models.

This result is somewhat surprising. The intuitive expectation underlying the design of this study was that an abliterated model — one whose safety-aligned refusal behavior has been surgically removed (2) — would function as a more capable adversarial attacker. Freed from the internal constraint of refusing harmful requests, the uncensored model should, in principle, generate more direct and aggressive adversarial content, and should more readily inhabit the adversarial framing required by the system prompt. The data contradicts this expectation.

A likely explanation lies in the collateral damage that abliteration inflicts on general model capability. Shairah et al. (2025) (1) directly studied this question and found that abliteration induces slight reductions in MMLU benchmark scores (0.5–1.3%) and moderate increases in perplexity. While these effects may appear negligible in isolation, they carry outsized consequences in the PAIR setting. Successful iterative red-teaming requires the attacker to perform nuanced reasoning across multiple turns: it must accurately diagnose why a prior attempt failed, select a qualitatively different strategy, and compose a coherent and rhetorically persuasive prompt that approaches the target's refusal threshold from a new angle. Even marginal degradation in instruction-following or language modeling precision compounds across 10 refinement iterations, eroding the attacker's ability to exploit the feedback signal that the PAIR loop provides.

4.2 RQ2: Semantic Strategy Analysis

What semantic strategies do the two attacker types employ, and how does strategy preference relate to strategy effectiveness?

4.2.1 Annotation Methodology

We adopt a semantic strategy taxonomy inspired by Liu et al. (2023) (5), who classify jailbreak prompts into broad strategies such as *Pretending*, *Attention Shifting*, and *Privilege Escalation*. We refined and adapted their categories to better reflect the strategy space observed in this experiment, our taxonomy consists of five categories: **Hypothetical Framing** (embedding the harmful request within a fictional or counterfactual scenario), **Persona Adoption** (instructing the target to assume an alternative, unconstrained identity), **Authority Roleplay** (impersonating an expert or institutional role to legitimize the request), **Obfuscation** (encoding or restructuring the request to evade surface-level pattern matching), and **Indirect Elicitation** (decomposing the harmful request into individually benign sub-tasks).

Assigning a semantic strategy label to each generated prompt is a non-trivial classification task: prompts frequently blend multiple rhetorical devices, and the boundary between categories such as Hypothetical Framing and Persona Adoption can be ambiguous in practice. To produce reliable labels at scale without prohibitive manual effort, we employ an automated majority-vote annotation system using three LLM judges: the two attacker models (`Llama-3-8B-Instruct-Abliterated` and `Meta-Llama-3-8B-Instruct`) and the target model (`Gemma-2-9B-it`). Each judge independently classifies a given prompt into one of the five taxonomy categories, and the final label is determined by majority vote across the three

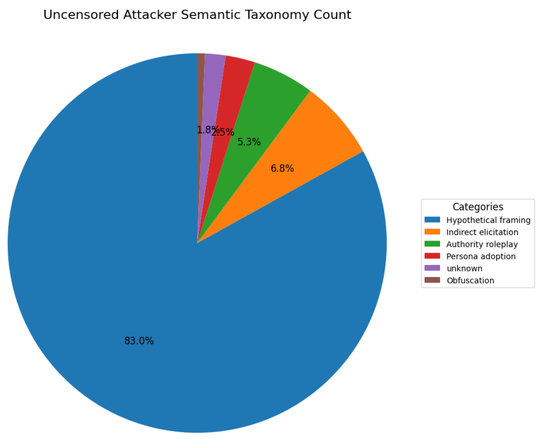

Figure 2: Strategy distribution across all prompts generated by the uncensored (abliterated) attacker.

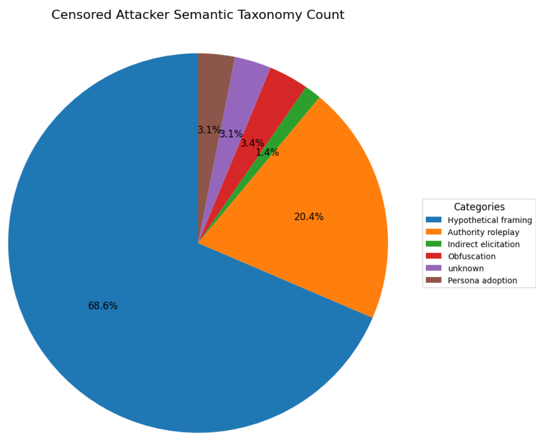

Figure 3: Strategy distribution across all prompts generated by the censored attacker.

judges. In cases where all three judges disagree, the prompt is assigned to the Unknown category, which accounts for the small unknown slices visible in the distribution figures.

To validate this automated scheme, we manually labeled a random sample of 20 prompts and compared our hand-assigned labels against the majority-vote outputs. The system achieved 85% agreement with human annotation, a level of alignment we consider sufficient for characterizing broad distributional trends across hundreds of prompts. Cases of disagreement were most frequently observed at the boundary between Hypothetical Framing and Authority Roleplay – prompts that adopt an expert persona within a fictional scenario. The 85% validation rate gives us confidence that the aggregate distributions reported below are reliable, while the residual disagreement rate motivates the Unknown category as an honest accounting of classification uncertainty rather than forcing ambiguous cases into a potentially incorrect label.

4.2.2 Overall Strategy Distribution

We first characterize each model's strategy distribution across *all* generated prompts, regardless of outcome. Despite sharing the same base architecture and system prompt, the two models exhibit meaningfully different preference profiles.

As seen in Figure 2, The uncensored attacker is overwhelmingly concentrated on Hypothetical Framing, which accounts for 83.0% of all generated prompts. The remaining strategies — Indirect Elicitation (6.8%), Authority Roleplay (5.3%), and Persona Adoption (2.5%) — each represent a marginal share, with Obfuscation near-absent. This near-singular reliance on one strategy suggests a narrow and relatively undifferentiated attack options: faced with any harmful behavior, the abliterated model consistently defaults to hypothetical framing with limited strategic variation across iterations or behavior categories.

The censored attacker presents a notably more distributed profile (Figure 3). While Hypothetical Framing remains dominant at 68.6%, Authority Roleplay constitutes a substantial 20.4% of all attempts — more than three times its share in the uncensored model's output. Obfuscation (3.4%), Unknown (3.1%), and Persona Adoption (3.1%) each appear at small but non-negligible rates. This broader distribution suggests that the censored model's generation tendencies are less concentrated than the abliterated model's, producing a more varied mix of strategies across the prompt pool even without any explicit instruction to diversify.

4.2.3 Strategy Distributions Among Successful Attacks

Restricting analysis to successful jailbreaks reveals a striking convergence between the two models. Both achieve success almost exclusively through *Hypothetical Framing*.

As shown in Figures 4 and 5,both models show remarkably similar success distributions despite their differing overall strategy profiles. The uncensored model's successful attempts break down as 81.8% Hypothetical Framing, 9.1% Authority Roleplay, and 9.1% Persona Adoption, while the censored model's are similarly concentrated at 84.6% Hypothetical Framing, 7.7% Authority Roleplay, 3.8% Persona Adoption, and 3.8% Unknown. Mir-

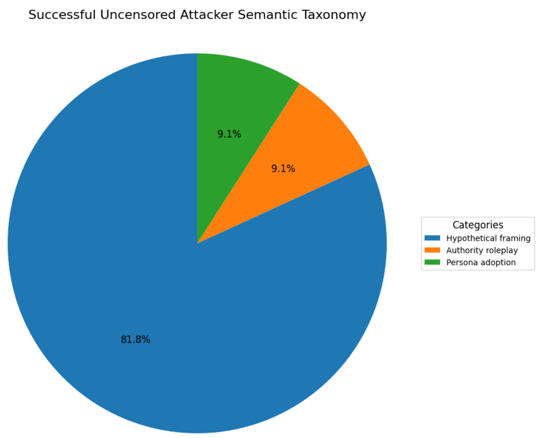

Figure 4: Strategy distribution restricted to successful jailbreaks by the uncensored attacker.

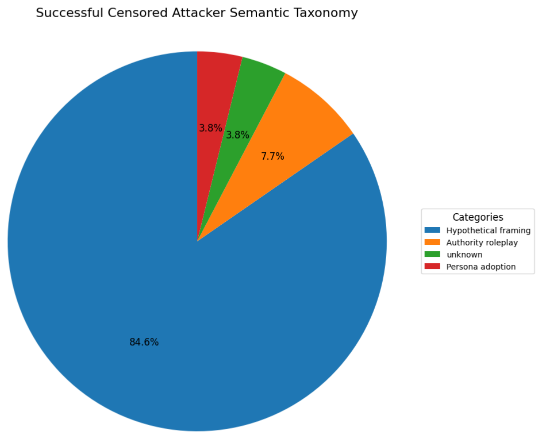

Figure 5: Strategy distribution restricted to successful jailbreaks by the censored attacker.

roring the pattern observed in the full taxonomy distributions, the vast majority of successful jailbreaks, regardless of attacker type, are achieved through Hypothetical Framing, reinforcing its role as the dominant attack strategy in this experimental setting.

4.2.4 Preference-over-Effectiveness Analysis

To formally characterize the relationship between strategy preference and per-attempt effectiveness, we compute a **Preference-over-Effectiveness (PoE)** ratio for each strategy s:

$$\text{PoE}_s = \frac{\text{Total Attempts}_s}{\text{Successful Jailbreaks}_s}$$

A lower PoE indicates a more efficient strategy, fewer total attempts are needed per success. A higher PoE indicates a strategy that is attempted frequently but converts at a low rate. Figure 6 displays the result.

The PoE results (Figure 6) reveal the most important quantitative finding of the strategy analysis: the two models differ dramatically in how efficiently they exploit Hypothetical Framing. The censored model achieves a PoE of 11.00 on this strategy, roughly 11 attempts per successful jailbreak. The uncensored model's PoE is 42.17, nearly four times higher, meaning that despite its overwhelming preference for this strategy, the abliterated model converts individual hypothetical framing attempts at a fraction of the censored model's rate. This discrepancy directly instantiates the instruction-following degradation account: the uncensored model attempts hypothetical framing at high frequency but generates less precisely calibrated instantiations of it, producing hypothetical frames that are either too obvious to bypass the safety classifier, or not sufficient enough to get a coherent malicious compliance response from the target.

For Authority Roleplay, the pattern partially inverts: the uncensored model achieves a PoE of 24.00 while the censored model's PoE is 36.00, suggesting that the abliterated model is comparatively more efficient per attempt at this strategy. This is an interesting exception — it may reflect that Authority Roleplay is a more formulaic strategy that requires less nuanced rhetorical calibration and is therefore less sensitive to the kind of reasoning degradation induced by abliteration. However, given that Authority Roleplay constitutes only 5.3% of the uncensored model's total attempts (compared to 20.4% for the censored model), this per-attempt efficiency advantage does not translate into meaningful absolute gains.

Obfuscation and Indirect Elicitation produced zero successful jailbreaks for both attacker types. The cause is unclear — it may reflect the target model's robustness to these strategies, the attacker models' limited proficiency at generating them, or both. Disambiguating these factors would require a more targeted experiment varying the target model and attacker independently.

4.2.5 Summary

The strategy analysis yields three findings. First, Hypothetical Framing dominates successful jailbreaks for both attacker types — whether this reflects a weakness in the target model, a generation bias in the attackers, or both remains an open question; RQ3 provides suggestive but not conclusive evidence for a target-side vulnerability. Second, the censored model exhibits a more varied overall

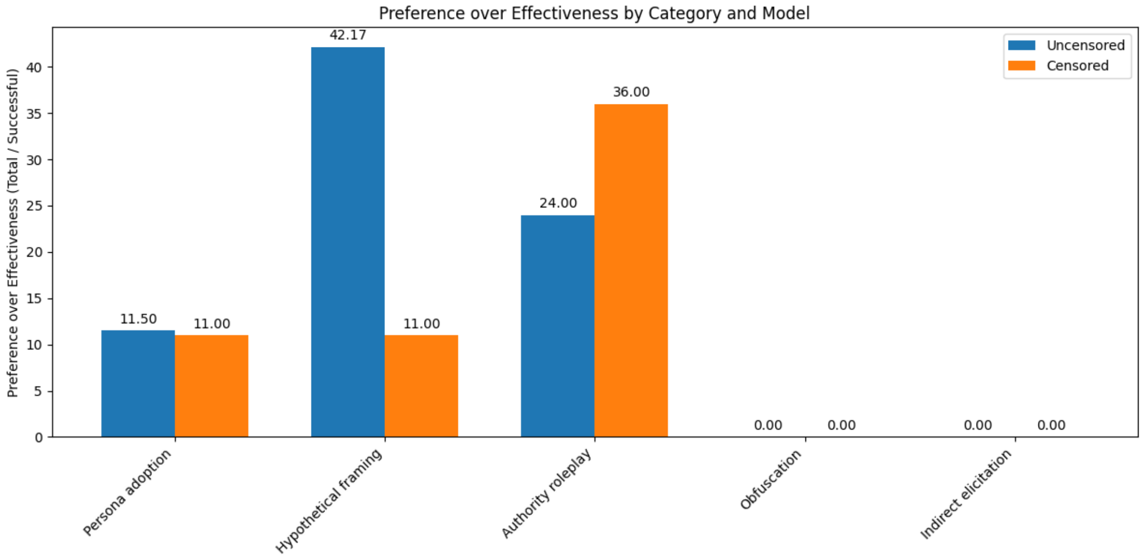

Figure 6: Preference-over-Effectiveness (PoE) by strategy. A lower PoE (fewer total attempts per success) indicates a more efficient strategy.

strategy distribution, though whether this reflects deliberate adaptation or simply greater generative diversity cannot be determined from aggregate data alone. Third, and most critically, the censored model's higher ASR is not a product of attempting more prompts, but of executing Hypothetical Framing with nearly four times greater per-attempt efficiency (PoE of 11.00 vs. 42.17) — a finding consistent with the instruction-following degradation account established in RQ1.

4.3 RQ3: Cross-Model Transferability

Do adversarial prompts transfer across target models of the same parameter count, and how does ASR change under transfer?

To evaluate cross-model generalization prior to assessing prompt transferability, we first establish baseline attack performance against the secondary target `Qwen3.5-9B`. We replicate the RQ1 experimental setup in its entirety — the same PAIR-based refinement loop, attacker models, and 50-behavior HarmBench sample — with `Qwen3.5-9B` substituted as the target in place of `Gemma-2-9B-it`. The abliterated attacker achieves an ASR of 0.32, while the censored attacker achieves 0.30, representing a notable reduction relative to performance against the primary target.

To evaluate transferability, we collected all successful adversarial prompts from both attacker models against the primary target (`Gemma-2-9B-it`), yielding a pool of 48 prompts (22 from the uncensored attacker, 26 from the censored attacker). These prompts were submitted directly to `Qwen3.5-9B` without any modification or iterative refinement — each prompt received up to 10 attempts, terminating early on a successful verdict from the judge classifier.

Attacker Model	ASR
Uncensored (Llama-3-8B-Abliterated)	0.32
Censored (Meta-Llama-3-8B-Instruct)	0.30

Table 3: Attack Success Rate (ASR) on Qwen3.5-9B for both attacker models.

The transfer experiment yielded a success rate of 61.2% across the transferred prompt pool. This figure is notably higher than both the original ASR values observed against `Gemma-2-9B-it` (Table 2), and the ASR values from performing the baseline experiment with `Qwen3.5-9B` (Table 3). This suggests that prompts optimized to bypass one aligned target do not merely transfer — they transfer with elevated effectiveness against a structurally different model of equivalent parameter count. This also suggests that jailbreaking an aligned model with better curated attacking prompts yields better results than attempting to do the same act via an attack and refine loop.

This result has two plausible interpretations. First, the successful prompts from RQ1 were already selected for their rhetorical effectiveness — they represent the top-performing outputs of an iterative refinement process, and as such carry an inherent quality advantage over the average prompt in the original attack loop. Second, and more substantively, the dominance of Hypothetical Framing among successful prompts (established in RQ2) may indicate that this strategy exploits a general vulnerability shared across aligned models of this scale, rather than a weakness specific to `Gemma-2-9B-it`'s safety training. If hypothetical reframing is effective because it places the target in a plausible deniability context, this mechanism is likely architecture-agnostic and would apply to any instruction-tuned model that was not explicitly trained to resist fictional framing.

5 Conclusion

This study set out to test the intuition that abliteration would yield a more capable adversarial attacker. Our results consistently contradict this expectation. Across all the research, the censored attacker closely matches or outperforms its abliterated counterpart, suggesting that the properties abliteration removes are not merely safety constraints but functional capabilities that underpin effective adversarial reasoning.

RQ1 established that the censored attacker achieves a higher ASR (0.52 vs. 0.44), a gap consistent with the instruction-following degradation that abliteration is known to induce (1). RQ2 deepened this account: while both attacker models heavily favor Hypothetical Framing, the censored model executes it with dramatically greater per-attempt efficiency (PoE of 11.00 vs. 42.17), and maintains a more strategically diversified repertoire of strategy overall. RQ3 further revealed that successful prompts transfer to a structurally different target (`Qwen3.5-9B`) at a 61.2% success rate — exceeding the original ASR against `Gemma-2-9B-it` despite `Qwen3.5-9B` exhibiting lower baseline susceptibility to direct iterative attack — pointing to Hypothetical Framing as a general vulnerability shared across aligned models of this parameter scale rather than a target-specific weakness.

Taken together, these findings carry practical implications for red-teaming pipeline design: abliterated models are not a reliable upgrade over their aligned counterparts as attacker components, and may in fact be counterproductive. Future work should examine whether these patterns hold across larger parameter scales, alternative base architectures, and attacker models fine-tuned specifically for adversarial prompt generation rather than general instruction-following.

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